

Statistical Inference & Sampling

DSAT Math Practice | 25 Questions | Difficulty-Graded | ~15-20% Grid-Ins

Directions: Answer each question by selecting the best answer from the choices given, or by entering your answer in the grid-in box for student-produced response questions. You may use a calculator on all questions. Reference the provided formulas as needed.

1

Easy

A researcher wants to estimate the average number of hours college students in the United States spend studying per week. Which of the following sampling methods would best allow the researcher to generalize results to all college students in the United States?

A

Survey 500 students at one large university.

B

Survey 200 students at a community college.

C

Select a random sample of 500 students from colleges across the United States.

D

Survey all 300 students in the researcher's introductory psychology class.

2

Easy

A biologist captures 40 fish from a lake, tags them, and releases them. Two weeks later, she captures 50 fish and finds that 8 of them are tagged. Which of the following is the best estimate of the total number of fish in the lake?

A

200

B

250

C

320

D

400

3

Easy

A survey of 600 registered voters in a city found that 54% support a new public transit initiative, with a margin of error of 4 percentage points. Which of the following is the best interpretation of this result?

A

Exactly 54% of all registered voters support the initiative.

B

Between 50% and 58% of the surveyed voters support the initiative.

C

It is plausible that the true proportion of all registered voters who support the initiative is between 50% and 58%.

D

The initiative will pass with 54% of the vote.

4

Easy

A company surveys its 50 newest customers about satisfaction. The results show that 80% are satisfied. Can the company conclude that 80% of all its customers are satisfied?

A

Yes, because the sample size is large enough.

B

Yes, because 80% is a high satisfaction rate.

C

No, because the sample is not representative of all customers.

D

No, because the margin of error is too large.

5

Easy

Which of the following is an example of a voluntary response sample?

A

A teacher randomly selects 10 students from her class list.

B

A news website asks visitors to click a button to participate in a poll.

C

A researcher uses a random number generator to select households.

D

Every fifth person entering a store is asked to fill out a survey.

6

Easy

A study finds that students who eat breakfast score higher on exams than those who do not. Which conclusion is most appropriate?

A

Eating breakfast causes higher exam scores.

B

There is an association between eating breakfast and higher exam scores.

C

Students should eat breakfast to improve their scores.

D

Not eating breakfast causes lower exam scores.

7

Medium

A polling organization surveys 1,200 likely voters and reports that Candidate A has 48% support with a margin of error of 3 percentage points at a 95% confidence level. Which of the following

conclusions is supported?

- A Candidate A will definitely lose the election.
- B Candidate A has between 45% and 51% support, so the race is too close to call.
- C Exactly 48% of all voters support Candidate A.
- D If the poll is repeated, exactly 95% of polls will show 48% support.

8

Medium

A researcher randomly assigns 120 participants to two groups. Group 1 receives a new medication, and Group 2 receives a placebo. After 8 weeks, Group 1 shows significantly lower blood pressure. What can the researcher conclude?

- A The medication is associated with lower blood pressure in the general population.
- B The medication caused the lower blood pressure in the participants.
- C Lower blood pressure causes people to take medication.
- D The placebo had no effect on blood pressure.

9

Medium

A national survey of 2,500 adults finds that 62% support renewable energy tax credits, with a margin of error of 2 percentage points. A state-level survey of 400 adults in State X finds 58% support. Which statement is most accurate?

- A State X has significantly less support than the national average.
- B The difference could be due to sampling variability, since 58% falls within the national margin of error.
- C The state survey is unreliable because it has fewer respondents.
- D The national and state surveys contradict each other.

10

Medium

A wildlife biologist uses the capture-recapture method. She tags 60 deer in a forest. Later, she captures 80 deer and finds that 12 are tagged. What is the estimated total deer population in the forest?

[Student-Produced Response]

11

Medium

A hospital wants to study the effectiveness of a new surgical technique. Which of the following study designs would allow the hospital to establish a causal relationship?

A

Observe patients who chose the new technique and compare outcomes.

B

Survey patients about their satisfaction with the technique.

C

Randomly assign eligible patients to the new technique or the standard technique.

D

Compare outcomes at hospitals that use the new technique with those that do not.

12

Medium

A sample of 800 households in a metropolitan area finds that the average monthly electricity bill is \$142 with a margin of error of \$8 at a 95% confidence level. Which of the following is the correct interpretation?

A

95% of households pay between \$134 and \$150.

B

We are 95% confident that the true average monthly bill for all households in the area is between \$134 and \$150.

C

The average bill will be between \$134 and \$150 in 95% of months.

D

There is a 95% probability that any individual household's bill is between \$134 and \$150.

13

Medium

A school district surveys parents about a proposed schedule change. Surveys are sent home with students, and 30% are returned. The returned surveys show 75% support for the change. Which of the following is the most significant concern about this result?

A

The sample size is too small.

B

The survey question may have been biased.

C

Parents who returned the survey may differ systematically from those who did not (nonresponse bias).

D

The margin of error is too large.

14

Medium

Two studies investigate the relationship between exercise and mental health. Study A is an observational study of 5,000 people. Study B is a randomized controlled experiment with 200 people. Both find that exercise is linked to better mental health. Which study provides stronger evidence of a causal relationship?

A

Study A, because it has a larger sample size.

B

Study B, because random assignment controls for confounding variables.

C

Both provide equally strong evidence.

D

Neither study can provide evidence of causation.

15

Medium

A poll of 900 adults found that 63% favor stricter environmental regulations. The margin of error is $1/\sqrt{n}$, where n is the sample size. What is the margin of error for this poll, rounded to the nearest hundredth?

[Student-Produced Response]

16

Medium

A researcher conducts a study and finds a statistically significant difference between two groups ($p\text{-value} < 0.05$). Which of the following is the best interpretation?

A

The difference between the groups is large and meaningful.

B

The observed difference is unlikely to have occurred by random chance alone.

C

There is a 95% chance the results are correct.

D

The experiment was well-designed.

17

Hard

A random sample of 400 employees at a large corporation finds that 45% are satisfied with their benefits package, with a margin of error of 5 percentage points. The company has 10,000 employees. If the company were to survey 1,600 employees instead, what would happen to the margin of error?

A

It would be cut in half (approximately 2.5%).

B

It would be reduced to one-quarter (approximately 1.25%).

C

It would remain at 5%.

D

It would increase because a larger sample has more variability.

18**Hard**

An agricultural researcher wants to test whether a new fertilizer increases crop yield. She divides 40 identical plots into two groups of 20. Group A receives the new fertilizer; Group B receives the standard fertilizer. After harvest, Group A's mean yield is 15% higher than Group B's, and the result is statistically significant. Which of the following is the most appropriate conclusion?

A

The new fertilizer causes a 15% increase in yield for all crops everywhere.

B

The new fertilizer caused a 15% increase in yield for the plots in this experiment, and the result is unlikely due to chance.

C

The 15% difference is too small to be meaningful.

D

The experiment should be repeated with more plots before any conclusion can be drawn.

19**Hard**

A city council wants to know residents' opinions on a proposed bike lane. Council members stand outside a popular cycling shop and survey 200 people. 82% of respondents support the bike lane. Why might this result overestimate the true level of support among all city residents?

A

The sample size is too small.

B

The survey location is likely to attract people who already favor cycling (selection bias).

C

The margin of error is too large.

D

82% is too high to be a valid survey result.

20**Hard**

A sample of 500 students found that 340 prefer online textbooks. What proportion of the sample prefers online textbooks? Express your answer as a decimal.

[Student-Produced Response]

21

Hard

A clinical trial tests Drug X against a placebo. The researchers report: "Patients in the Drug X group had a 30% reduction in symptoms compared to placebo ($p = 0.03$).\" Which of the following interpretations is correct?

A

There is a 3% chance that Drug X does not work.

B

There is a 97% probability that Drug X is effective.

C

If Drug X had no real effect, there would be only a 3% probability of observing a difference this large or larger by chance.

D

Drug X works for 97% of patients.

22

Hard

A study of 10,000 people finds a statistically significant correlation between coffee consumption and longevity ($p < 0.001$). However, the actual difference in lifespan is 0.3 months. Which of the following best describes this finding?

A

The result is both statistically and practically significant.

B

The result is statistically significant but may not be practically significant.

C

The result is practically significant but not statistically significant.

D

The result is neither statistically nor practically significant.

23

Hard

A marketing firm conducts a survey about brand preference. They call landline phones between 10 AM and 2 PM on weekdays. Which demographic group is most likely to be underrepresented in this survey?

A

Retired individuals

B

Stay-at-home parents

C

Working adults who are at their jobs during these hours

D

People who live in rural areas

24

Hard

A pollster reports that a candidate has 52% support with a 95% confidence interval of (49%, 55%). If the election requires more than 50% to win, can the pollster conclude with 95% confidence that the candidate will win?

[Student-Produced Response]

25

Hard

Researchers want to determine whether a tutoring program improves test scores. They compare test scores of students who voluntarily enrolled in tutoring with scores of students who did not. The tutored students scored higher. A critic argues this does not prove the program works. Which of the following best supports the critic's argument?

A

The sample size was too small.

B

Students who chose tutoring may have been more motivated, creating a confounding variable.

C

The test was not standardized.

D

The researchers did not calculate a p-value.
